

Minimize Pressure Difference Traffic Signal Control Based on Deep Reinforcement Learning

Pengcheng Yu, Jie Luo

College of Automation & College of Artificial Intelligence, Nanjing University of Posts and Telecommunications, Nanjing 210023, China
E-mail: ljnc@njupt.edu.cn, pc.yu@foxmail.com

Abstract: In the traffic signal light control method based on deep reinforcement learning, there is a challenging problem to coordinate control traffic signal because of the high complexity of the road network. This paper proposes a minimize pressure difference traffic signal control based on deep reinforcement learning. It use traffic pressure difference at each phase of the intersection as the reward. The pressure value considers the queue length of vehicles and the capacity of the roads. The traffic state of the intersection is converted into a two-dimensional matrix composed of vehicle position and speed information, the control strategy is learned online through the deep double Q-learning network model to realize the adaptive control of the traffic signal. The algorithm proposed in this paper is simulated by the traffic simulation platform SUMO and compared with two existing traffic signal control methods based on deep reinforcement learning. The experimental results show that the algorithm has a faster convergence rate and better performance in regional traffic signal coordination control.

Key Words: Adaptive Traffic Signal Control, Multi-agent System, Deep Reinforcement Learning; Pressure Difference

1 Introduction

With the acceleration of urbanization, urban traffic congestion has become a serious social problem. Traffic signal control systems can alleviate traffic congestion. In traditional traffic control methods, such as the timed traffic control method and the induction signal control method, it is difficult to handle dynamic traffic conditions well. However, the existing control systems such as SCOOT [1] and SCATS [2] have insufficient dynamic adaptability to complex traffic conditions.

With the rapid development of artificial intelligence and transportation Internet of Things technology, traffic signal control based on reinforcement learning (RL) has become a hot issue in intelligent transportation [3]. The RL [4] obtains the optimal control strategy through interactive learning between the agent and the environment. The RL is applied to traffic signal control [5], which adaptively learns the optimal control strategy according to the traffic conditions signalling strategies to improve traffic conditions.

A key question for reinforcement learning to traffic signal control is how to define the state, reward and action. Recent studies [6, 7] usually define the reward as a weighted linear combination of several other traffic states. For a comprehensive description of traffic conditions, some methods use complex state definitions, such as using visual images [7, 8] to represent states, which leads to the curse of dimensionality problem and increases the learning time.

In the research of regional traffic signal control based on reinforcement learning. One way is to use the centralized control method, each agent controls an intersection, and the behaviour between the agents is jointly modelled and learned [9, 10]. Since these methods require negotiation among all agents in the entire network, Algorithms are computationally complex. Another way uses decentralized RL agents to control traffic signals [11–14]. Each agent makes its own decision by using the information of itself and its neighbours. The distributed method is scalable and practical, the method proposed in this paper also follows this direction.

This paper proposes a traffic signal control method based

on multi-agent deep reinforcement learning (MARL). In the regional traffic signal control system, each control node in the regional road network is used as an agent to jointly participate in coordinating and optimising the global traffic network flow. The agent uses the real-time traffic information to discretise the state encoding as the input of the neural network, which improves the accuracy of the traffic state representation and avoids the dimension disaster problem caused by the large state space. Using the double DQN (DDQN) [15] as the algorithm model, each agent uses the traffic pressure difference as the reward. The differential pressure value represents the degree of imbalance between the density of vehicles in the incoming and outgoing of the intersection. Store experiences to replay memory, select mini-batch samples using priority experience sampling strategy, the control strategy is updated to minimise the pressure difference to realise the coordinated control of multi-junction traffic and improve the traffic efficiency of the road network.

2 Multi-Agent Deep Reinforcement Learning

2.1 Road Network Model

This paper studies the dynamic control problem of the traffic light phase in multi-intersection. As shown in Fig. 1, an intersection has four directions (east, west, south, north), divided into three lanes (go straight lane, left-turn lane, right-turn lane) and eight possible phases. One legal design for stages is east-west straight green light (EWG), east-west left turn green light (EWLG), north-south straight green light (SNG), north-south left turn green light (SNLG). Use σ represent phase motion sets as

$$\sigma = \{(\phi_1, \phi_2), (\phi_3, \phi_4), (\phi_5, \phi_6), (\phi_7, \phi_8)\} \quad (1)$$

2.2 Deep Reinforcement Learning

In the DQN proposed by DeepMind [16], DQN integrates the CNN and the Q-learning algorithm. The algorithm establishes a neural network model and trains the CNN parameters through the obtained experience. According to the input of the state, the corresponding Q value is directly output.

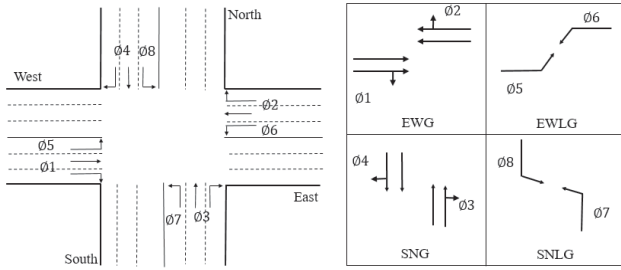


Fig. 1: Phase and traffic movements in traffic signal control

To train the value function, the algorithm needs the action-value (also known as Q) function and the target value function. The introduction of the target values is to provide an approximation to the action-values. The action-values $Q(s, a; \theta)$ and the target values $Q(s, a; \theta^-)$ of the current state can be expressed as the output of the CNN, where θ and θ^- are the network parameters. Use a deep convolutional neural network to approximate the optimal action-value function

$$Y_t^{DDQN} = R_{t+1} + \gamma \max_{a'} Q(s_{t+1}, a'; \theta_t^-) \quad (2)$$

Where R_{t+1} is the reward after performing the action; $\max_{a'} Q(s_{t+1}, a'; \theta_t^-)$ is the Q value corresponding to action a' in the next state s_{t+1} , γ is the discount factor.

Formula (2) selects the maximum Q value as the estimate of the action-values in selecting and evaluating the greedy behaviour, which leads to an overestimation problem. To avoid this problem, Van Hasselt proposed a DDQN [15], the maximum Q value corresponding to each action is not directly obtained in the target values network. First, the action corresponding to the maximum Q value is calculated through the action-values network. The network parameter is θ and the action a as

$$a' = \max_{a'} Q(s_{t+1}, a'; \theta_t) \quad (3)$$

Then use the selected action to calculate the target values in the target network to evaluate the action, and the corresponding network parameter is θ^- . The calculation formula of the target values of the DDQN can be expressed as

$$Y_t^{DDQN} = R_{t+1} + \gamma Q(s_{t+1}, \arg \max_{a'} Q(s_{t+1}, a'; \theta_t), \theta_t^-) \quad (4)$$

The update of the parameter θ by the Q value in the algorithm calculates the gradient of the loss function through the gradient descent method, and the loss function $L(\theta)$ is calculated using the root mean square error formula

$$L(\theta_t) = E \left[\left(R_{t+1} + \gamma Q(s_{t+1}, \arg \max_{a'} Q(s_{t+1}, a'; \theta_t), \theta_t^-) - Q(s_t, a; \theta_t) \right)^2 \right] \quad (5)$$

Where Y^{DDQN} is the target values of the Q network in DDQN; $\arg \max_{a'} Q(s_{t+1}, a; \theta_t)$ is the action corresponding to the highest action-values output by the Q network in

the state s_{t+1} , $Q(s_{t+1}, \arg \max_{a'} Q(s_{t+1}, a'; \theta_t); \theta^-)$ is the action-values output by the target network according to the action selected by the Q network in the state s_{t+1} ; γ is the discount factor.

This paper adopts a DDQN algorithm based on prioritized experience replay (PER) [17], replacing uniform sampling with priority sampling. In the training process, The agent selects action a_t in the current state s_t according to the greedy strategy, get the reward r_t and the next state s_{t+1} . A sample e_t is defined as the quadruple $e_t = (s_t, a_t, r_t, s_{t+1})$. For the samples in the replay memory, the difference in importance of the samples is considered, and the more important or valuable experience is given a higher probability of being sampled.

2.3 Minimizing Pressure Difference Traffic Signal Control Based on Deep Reinforcement Learning

The regional traffic signal control method based on RL belongs to MARL. It is not feasible to train a centralized single-agent control method. With the increase in the number of intersections, the joint action space grows exponentially. In a multi-intersection traffic signal control method based on independent agent reinforcement learning, Each intersection as an agent, and each agent independently perceives the local environment around the intersection. The agent selects a signal adjustment scheme according to the current traffic state to achieve traffic signal control.

Based on the above scenarios, this paper proposes a normalized pressure difference method as reward. The pressure difference represents the degree of imbalance between vehicle densities on the incoming and outgoing lanes of the intersection. The goal of the policy update is to minimize the pressure difference. Use DDQN to select and evaluate agent actions to avoid overestimation. During the policy update process, each agent pools the acquired experience in replay memory to maximize the sum of the Q value functions. Use PER to select mini-batch samples to update the policy. Through the continuous iteration of training, the agent can choose the best signal control strategy according to different states and realize the adaptive control of traffic signals.

3 Traffic Control Method Based on Multi-Agent Deep Reinforcement Learning

3.1 State

This paper use the discrete traffic state encoding (DTSE) method to divide the lane into multiple cells with a fixed length and takes it as the input state. This method can accurately reflect the real-time position and speed information of vehicles near the intersection and can also be used to predict the subsequent traffic state.

As shown in Fig. 2, within the distance from each incoming lane to the stop line, divide each incoming lane into a cell with a fixed interval. According to the road speed limit value, the average speed of vehicles in the cell is normalized, get the position information and speed information of vehicles. v_{n_i} is the normalized speed of the vehicles

$$v_{n_i} = \frac{v_i}{v_{max}} \quad (6)$$

Where v_i is the actual speed, v_{max} is the maximum speed

allowed by the road; Fig. 2 shows the corresponding relationship between the simulated vehicle at the intersection in SUMO and the position matrix and the normalized speed matrix.

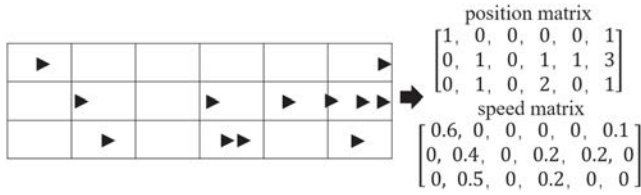


Fig. 2: Matrix representation of traffic states

3.2 Action

After the agent obtains the traffic state of the current intersection, it will select an action to execute in the action set. The change of the phase should be periodic to avoid traffic chaos caused by irregular switching between different phases. In this traffic signal control, the phase cycle set is as shown in formula (1), and the phases are transformed in the order of east-west straight, east-west left turn, north-south straight, north-south left turn, and the agent's action set $A = \{1, 0\}$. At time t , the state of the intersection is s_t , agent selects the action $a_t \in A$, where $a_t = 0$, the intersection keeps the current phase unchanged; $a_t = 1$, it switches to the next phase of the current phase in the phase sequence with a shortest phase duration of $\Delta t = 5s$. Set 3s yellow light time for clearing vehicles that have not yet driven out of the intersection.

3.3 Reward

A reward is an evaluation of a signal control decision and will influence the next signal control decision. This paper proposes a lane normalized pressure difference calculation method and the sum of pressure difference values as reward. The pressure value represents the density of traffic flow on the road, and the pressure difference represents the degree of unbalance between the density of vehicles on the incoming and outgoing lanes of the intersection. The calculation method of normalized pressure proposed in this paper shows that the pressure increase is linear under low traffic load. With the increase in queue length, the magnitude of pressure increases exponentially. The greater the pressure value to get the green light said, the current phase longer the time, the better able to alleviate the traffic pressure. The basic form of the normalized pressure formula can be defined as

$$\rho_a = k \times Q_a + \omega \times (Q_a)^m \quad (7)$$

Where $\rho_a \in [0, 1]$ is the normalized pressure on the lane, 0 means there is no vehicle queue, and the pressure value is zero; When the queue length reaches the maximum capacity of the road, the pressure value $\rho_a = 1$, means that the road cannot receive vehicles from the upstream road. Q_a is the number of vehicles queued on lane a . k ensures that the Q_a curve is linear for small values, and $\omega \times (Q_a)^m$ make the Q_a curve grows exponentially as the queue length increases for large values. Assuming that C_a is the maximum capacity in in road a , when $Q_a = C_a$, $\rho_a = 1$, the parameters ω can be described as

$$\omega = \frac{1 - k \times Q_a}{(C_a)^m} \quad (8)$$

Fig. 3 assume that the maximum capacity of a lane is $C_a = 200$, take the normalized pressure function diagram when $k = 0.001$, $m = 4$, When the queue length $Q_a < 40$, the pressure value is a linear growth process, and when $Q_a > 40$, the pressure value increases exponentially.

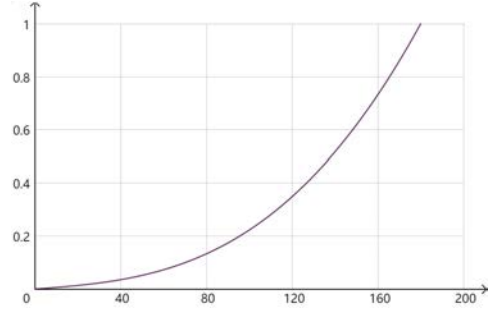


Fig. 3: Normalized pressure curve

The pressure difference is defined as the pressure difference value between incoming and outgoing lane. pressure difference of the traffic motion (a, b) of a vehicle from lane a into lane b as

$$\rho_{(a,b)} = \rho_a - \rho_b \quad (9)$$

Where $\rho_{(a,b)}$ is the pressure value on road a to road b . At the intersection i , the pressure value P_i at the intersection i is the sum of the pressures of the four phase movements as

$$P_i = \left| \sum_{(a,b) \in \sigma} \rho_{(a,b)} \right| \quad (10)$$

where σ is the phase motion set, as defined in formula (1). The reward r_i is shown in the following equation

$$r_i = -P_i \quad (11)$$

3.4 Deep Reinforcement Learning Network Model

The neural network structure of the algorithm is shown in Fig. 4, designed as a two-layer convolutional layer and a three-layer fully connected layer. The input of the neural network is a matrix composed of vehicle positions and speeds of size $L \times W \times 2$, and L and W represent the length and width of the state matrix. 2 is two eigenvalues of vehicle position information and vehicle speed. Extract high-dimensional features of intersection location and velocity distribution through two convolutional layers. After each convolution layer, maximum pooling is used for feature selection to reduce the number of parameters. After that, the high-dimensional feature vector is expanded into a one-dimensional vector as the state feature is extracted from the two sets of matrices, and then through the fully connected layer, the value distribution of the state main value function is finally output.

The training process of the Minimize Pressure Difference Traffic Signal Control Based on Deep Reinforcement Learning (MPD-DRL) can be shown in Algorithm 1.

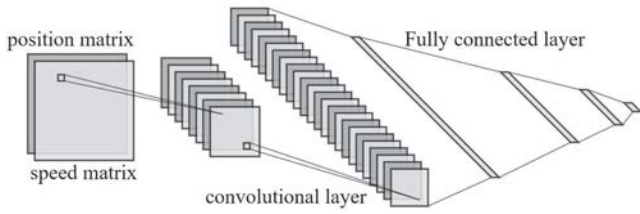


Fig. 4: Deep Neural Network Structure Diagram

Algorithm 1: MPD-DRL: Minimize Pressure Difference Traffic Signal Control Based on Deep Reinforcement Learning

Input: replay memory M , sample size B , greedy ε , discount factor γ , learning rate α , episode N , episode length T , replacement frequency C ;
Initialize network parameters θ, θ^- ; environment parameters, action set A
for each episode do
 Initialize step $i = 0$
 Get state s_t from traffic environment
 while $i \leq T$ **do**
 Select action a_t with ε -greedy policy
 Execute actions a_t , observe rewards r_t and next state s_{t+1}
 Store (s_t, a_t, r_t, s_{t+1}) in M
 Define the current traffic state $s = s_{t+1}$
 $i = i + 1$
 if $|M| > B$ **then**
 Sample B samples from M
 end
 Calculate the loss $L(\theta)$ by formula (5) and Update network parameters θ by perform a gradient decent step on loss function
 Update $\theta^- \leftarrow \theta$ every C step
 end
end

4 Simulation Experiments and Analyse

4.1 Simulation Environment and Parameter Settings

The simulation experiment adopts the Simulation of Urban Mobility (SUMO) traffic simulation platform. This paper takes a traffic area in Qixia District, Nanjing as the analysis object of the simulation experiment, as shown in Fig. 5, uses OpenStreetMap to select the regional road network as the research object, and uses SUMO's NetEdit to generate the regional traffic road network model, including 29 traffic lights.

During the running of the simulator, a certain number of vehicles will be generated at each time step and randomly distributed in the road network. Since the traffic flow at urban road intersections changes dynamically in each time period, in order to simulate the real traffic situation, this paper introduces a variable flow setting to simulate the real peak traffic flow scene. The flow changes are shown in Fig. 6. The x-axis is the time of each episode in the training and testing phases, and the episode length is set as 3,600 seconds, simulating the one-hour traffic control process in real traffic. The y-axis is the number of input traffic flows per hour, simulating the real road peak traffic conditions. The flow changes from small to large to reach the peak and gradually decreases

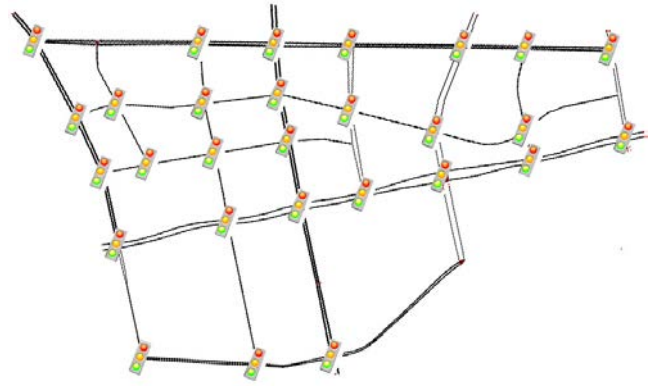


Fig. 5: Traffic network simulated in SUMO

after maintaining a specific peak flow. Compared with fixed traffic scenarios, variable traffic scenarios have a more complex state space and higher requirements for policy learning capabilities.

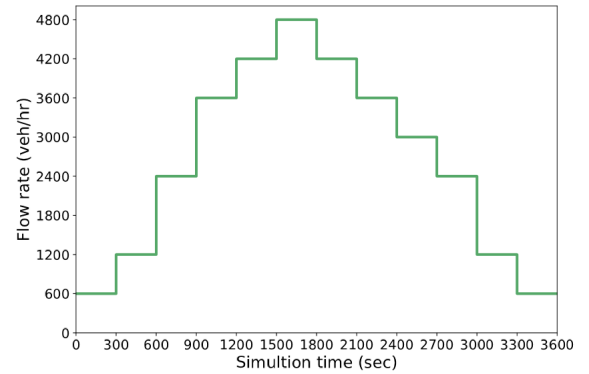


Fig. 6: Traffic flow generation distribution over one episode

4.2 Experimental Results and Analysis

To evaluate the effectiveness of the proposed algorithm, compared with the following two deep reinforcement learning algorithms. The hyper-parameters of network are listed in Table 1.

Table 1: Hyper-parameter settings

Parameter Type	Value
Replay memory (M)	10000
Mini batch size (B)	64
Learning rate (α)	10^{-4}
Reward discount factor (γ)	0.99
Initial greedy factor(ε)	0.9
Minimum greedy factor(ε)	0.05
Target network update rate	0.001
Episode (N)	1400
Episode length (T)	720
Control step	5s

(1)DQL [8]: A traffic signal control method based on deep Q learning. This method takes queue length as the definition of RL status and reward, regardless of traffic conditions at adjacent intersections. The algorithm adopts the same network structure and Hyperparameters setting as MPD-DRL.

(2)PressLight [18]: In order to verify the effectiveness and superiority of the proposed algorithm, the DDQN is adopted

as the network model of the algorithm. Take the queue length as the definition of the state, and take the difference between the number of vehicles on the incoming and outgoing lane as the reward, pressure value is linear growth.

The three algorithms are simulated and trained in the same simulation environment. All algorithms are trained with 1,400 episodes and each episode $T = 720$ steps. The data are plotted with scatter points, as shown in Fig. 7. The x-axis is the number of training steps, and the y-axis is each training episode average reward value.

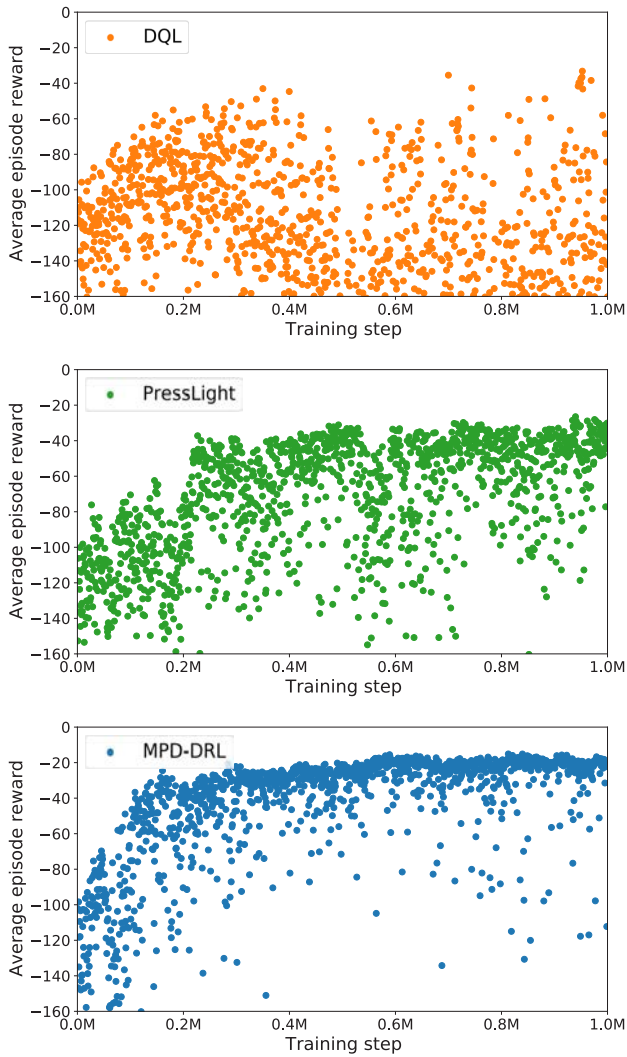
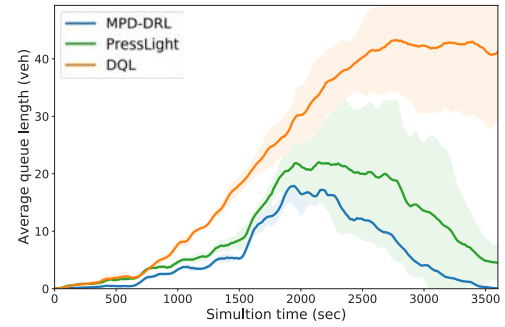


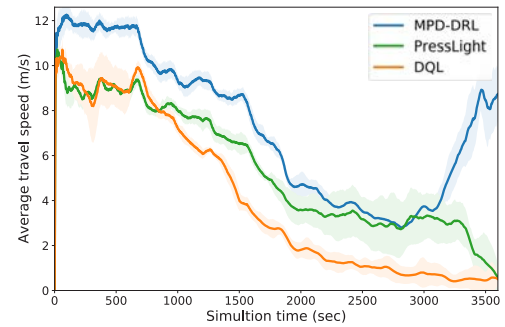
Fig. 7: Average cumulative reward value of three learning methods

DQL adopts the DQN algorithms, it uses the queue length as the definition of state and reward. The simple definition causes the algorithm learns an invalid control policy. MPD-DRL and PressLight algorithms lack experience samples in the initial learning phase, so they tend to choose random action strategies, resulting in algorithms that always get lower reward values. As the number of training rounds increases, the algorithm is trained by a large number of experience samples, the average reward value increases continuously and finally tends to converge, the algorithm learns an effective traffic coordination control strategy. Compared with PressLight, the MPD-DRL algorithm has faster convergence

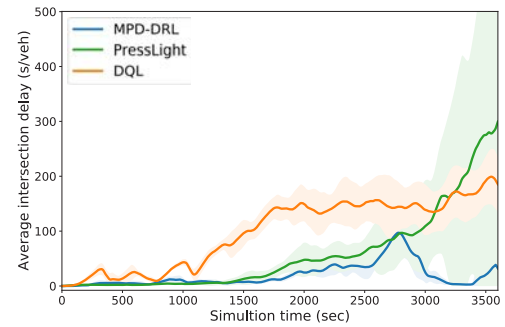
speed and lower dispersion of the average reward value, indicating better convergence and stability in the later stage of training.



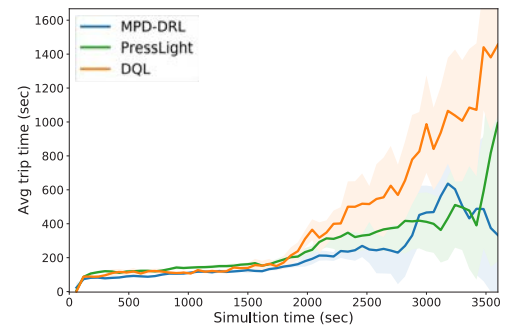
(a) Average queued length



(b) Average travel speed



(c) Average intersection delay



(d) Average trip time

Fig. 8: Comparison of different algorithm

In the three methods of traffic control made 10 times simulation test. Table 2 summarizes the average values of the key indicators compared in the evaluation. From the table, MPD-DRL algorithm has the best performance, and DQL is the worst. Compared with the PressLight algorithm, the MPD-DRL average travel time is reduced by 18.6%, the av-

Table 2: Performance of three methods

Algorithm	Average Travel Time	Average Travel Speed	Average Waiting Time	Average Queue Length	Average Intersection Delay
DQL	485	2.07	207	23.90	93.7
PressLight	237	5.04	157	10.45	51.4
MPD-DRL	193	5.96	114	6.73	36.8

erage travel speed is increased by 18.3%, the average delay time is reduced by 27.4%, the average queue length is reduced by 35.6%, and the average delay of vehicles at the intersection is reduced by 28.4%. The data show that the MPD-DRL method proposed in this paper can relieve the traffic pressure and achieve better traffic signal control.

As shown in Fig. 8, the average value of data obtained from 10 simulation experiments be plotted with different colors, the solid line part is the average of 10 test results, the shaded part is the standard deviation of the 10 experimental data, and the discrete interval of the data is obtained according to the standard deviation. The simulation experiment simulates the situation of peak traffic. With the increase of traffic flow, the road load and traffic pressure will increase.

Since the DQL algorithm did not converge in training, the intersection agent could not obtain an effective control strategy according to the traffic state, resulting in regional traffic congestion all the time. Although PressLight can realize the control of traffic signals after reaching the peak flow, the average travel speed and average travel time have not improved significantly, and the average delay at the intersection has been increasing, which shows that the algorithm cannot better realize the control of the traffic signal under the peak flow. Coordinated control of traffic signals. MPD-DRL has the best effect among the three control algorithms, which can better realize the coordinated control of adjacent intersections, distribute the traffic flow more evenly among the regional road networks, effectively reduce the average queuing length of vehicles and average travel time, which can better alleviate traffic congestion during peak hours.

5 Conclusion

This paper proposed a traffic signal control method based on normalized pressure difference deep reinforcement learning. Simulation experiments are carried out using actual intersectionS and flow data. Compared with two typical deep reinforcement learning algorithms, it can also get better performance in signals control. In the future, study the performance of traffic signal control under different definitions of states and rewards. As the number of intersections increases, coordination among multi-agents needs to be considered.

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